

7. State Output

The sensor is always sending cyclically (once per second) the current configuration and sensor state in message **RadarState** (0x201) and current firmware version in message **VersionID** (0x700).

For the filter configuration, the sensor only sends the current configuration as a reply to the filter configuration message **FilterCfg** (0x202). The sensor replies with the message **FilterState_Header** (0x203) with the number of configured filters and one message **FilterState_Cfg** (0x204) for the filter that has been configured.

When collision detection is active, additionally, the sensor sends the current collision detection configuration and warning state in message **CollDetState** (0x408) and for the individual regions in messages **CollDetRegionState** (0x402). Furthermore, the relay control message **CollDetRelayCtrl** (0x8) is sent, if it is activated with **RadarCfg_CtrlRelay**. They are also sent cyclically (once per second).

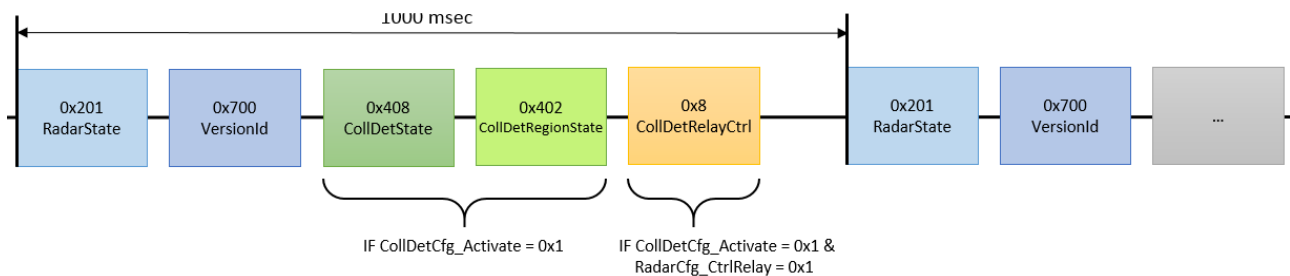


Figure 10: Overview of state messages that are sent cyclically by the radar

7.1. Radar state (0x201)

The message **RadarState** (0x201) is sent by the sensor in a regular interval (once per second). After configuring a radar configuration parameter, the corresponding signal in message 0x201 can be checked in order to verify that the configuration change has been accepted.

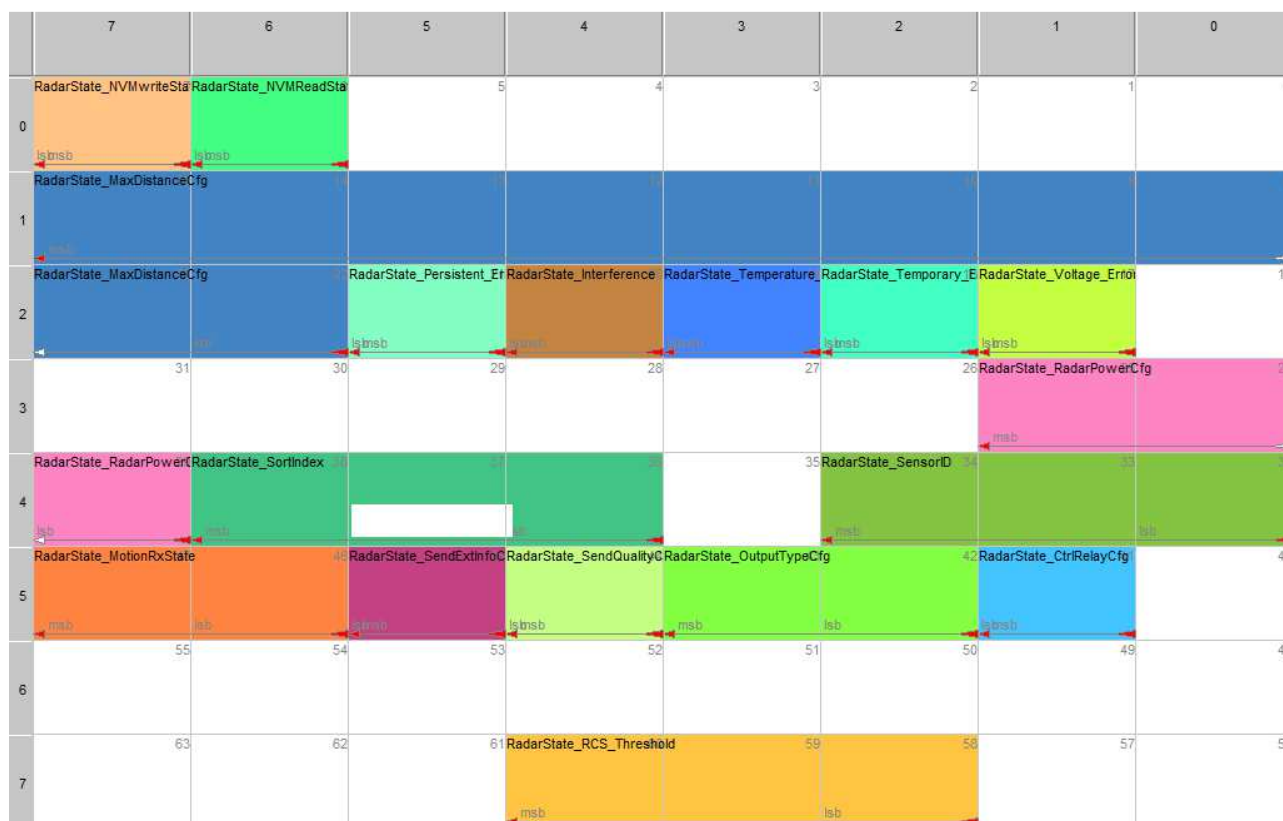


Figure 11: RadarState - message layout (0x201)

Table 10: RadarState - message content (0x201)

Signal	Start	Len	Min	Max	Res	Unit
RadarState_NVMMReadStatus	6	1	0	1	1	0x0: failed 0x1: successful
RadarState_NVMMwriteStatus	7	1	0	1	1	0x0: failed 0x1: successful

RadarState_MaxDistanceCfg	22	10	0	2046		ARS408: - Standard Range: 2 196 m – 260 m - Extended Range: 2 196 m – 1200 m ARS404: - Standard Range: 2 150 m – 190 m - Extended Range: 2 90 m – 1000 m
RadarState_Persistent_Error	21	1	0	1	1	0x0: No error 0x1: Persistent error active
RadarState_Interference	20	1	0	1	1	0x0: No interference 0x1: Interference detected
RadarState_Temperature_Error	19	1	0	1	1	0x0: No error 0x1: Temperature error active
RadarState_Temporary_Error	18	1	0	1	1	0x0: No error 0x1: Temporary error active
RadarState_Voltage_Error	17	1	0	1	1	0x0: No error 0x1: Voltage error active
RadarState_SensorID	32	3	0	7	1	Current sensor ID (0-7)
RadarState_SortIndex	36	3	0	7	1	0x0: no sorting 0x1: sorted by range 0x2: sorted by RCS
RadarState_RadarPowerCfg	39	3	0	7	1	0x0: Standard 0x1: -3dB Tx gain 0x2: -6dB Tx gain 0x3: -9dB Tx gain
RadarState_CtrlRelayCfg	41	1	0	1	1	0x0: inactive 0x1: active
RadarState_OutputTypeCfg	42	2	0	3	1	0x0: none 0x1: send objects 0x2: send clusters
RadarState_SendQualityCfg	44	1	0	1	1	0x0: inactive 0x1: active
RadarState_SendExtInfoCfg	45	1	0	1	1	0x0: inactive 0x1: active
RadarState_MotionRxState	46	2	0	3	1	0x0: input ok 0x1: speed missing 0x2: yaw rate missing 0x3: speed and yaw rate missing
RadarState_RCS_Threshold	58	3	0	7	1	0x0 Standard 0x1 High sensitivity

Table 11: RadarState - signal description (0x201)

Start	Signal	Description
6	RadarState_NVMReadStatus	State of reading the configuration parameters from non-volatile memory at startup
7	RadarState_NVMwriteStatus	State of storing a configuration parameter to non-volatile memory (initially this value is set to 0x0 and set to 0x1 after a configuration has been sent and successfully stored)
17	RadarState_Voltage_Error	Error will be active if the operating voltage is below or above the defined range for more than 5 seconds. U1 = < 7.1 VDC ---> Radar switches off U2 = 7.3-7.4 VDC ---> voltage error = 1 U3 = 7.5-7.6 VDC ---> voltage error = 0 Radar works U4 = 7.6-32.8 VDC---> Radars sends telegram U5 = 32.8 VDC ---> voltage error = 0 U6 = 32.9 VDC ---> voltage error = 1 U7 = > 32,9 VDC ---> Radar switches off It does always automatically quit after it has gone.
18	RadarState_Temporary_Error	A temporary error which will most probably disappear after a sensor reset has been detected.
19	RadarState_Temperature_Error	Error will be active if the temperature is below or above the defined range. It does automatically quit after it has gone.
20	RadarState_Interference	Interference with another radar sensor has been detected. In case of Interference there is no change in any interface function of the radar. Internally the background noise increases which decreases the performance slowly depending on the strength of interference, but the radar will go on in sending Clusters or Objects to the interface as long as any obstacle is detected. It does always automatically quit after it has gone. It should be used as an indication that the measurement of the radar is influenced by another radar and the performance will probably decrease a little bit. Every radar is protected against the effect of interference by a randomly unbalanced length of interrupts between the measurements. This avoids a permanent interference with another radar and it only creates partly interference with the assurance to leave immediately after appearing. To create full radar performance any interference should be avoided / kept as less as possible !!
21	RadarState_Persistent_Error	An internal error which might not disappear after a reset has been detected.
22	RadarState_MaxDistanceCfg	Current configuration of maximum distance of far scan – different of ARS 404 and ARS 408
32	RadarState_SensorID	Sensor ID 0 - 7
36	RadarState_SortIndex	Current configuration of sorting index for object list

39	RadarState_RadarPowerCfg	Current configuration of transmitted radar power parameter
41	RadarState_CtrlRelayCfg	True if relay control message is sent
42	RadarState_OutputTypeCfg	Currently selected output type as either clusters or objects
44	RadarState_SendQualityCfg	True if quality information is sent for clusters or objects
45	RadarState_SendExtInfoCfg	True if extended information is sent for objects
46	RadarState_MotionRxState	Shows the state of the speed and yaw rate input signals
58	RadarState_RCS_Threshold	If true the sensors high sensitivity mode is active